Electrochimica Acta 56 (2011) 4540–4543

Contents lists available at ScienceDirect

Electrochimica Acta

journal homepage: www.elsevier.com/locate/electacta

In situ electrochemical oxidation of ethylenediamine on Co–B alloy electrode during cycling for improving its electrochemical properties at elevated temperature

D.S. Lu[a,b,*], W.S. Li[a,b,*], C.L. Tan[a,b], Q.M. Huang[a,b]

[a] *Institute of Chemistry and Environment, South China Normal University, Guangzhou 510006, China*
[b] *Key Lab of Technology on Electrochemical Energy Storage and Power Generation in Guangdong Universities, Guangzhou 510006, China*

ARTICLE INFO

Article history:
Received 16 December 2010
Received in revised form 17 February 2011
Accepted 17 February 2011
Available online 24 February 2011

Keywords:
Cobalt-boron alloy electrode
Ethylenediamine
Cyclability
Electrochemical co-oxidation
Elevated temperature

ABSTRACT

A Co–B alloy/ethylenediamine (EDA) hybrid electrode system has been developed by adding EDA into electrolyte. Charge–discharge measurements show that the inorganic–organic hybrid electrode system exhibits high discharge capacity and long cycle life at elevated temperature (55 °C). Specifically, in the electrolyte containing 0.09 M EDA additive, the discharge capacity of Co–B alloy electrode after 100 cycles is still up to 601.7 m Ah g^{-1} at 55 °C. However, for the electrode in EDA-free electrolyte, its discharge capacity is sharply decreased to only 138.5 m Ah g^{-1} after 100 cycles. ICP-OES and IR measurements are used to clarify the reason of the improvement in the electrochemical properties. These results show that beneficial effect of EDA on electrochemical properties of Co–B alloy electrode can be attributed to *in situ* electrochemical oxidation of EDA during discharge cycles. First the oxidation of EDA contributes to part of the discharge capacity and secondly the oxidation products absorbed on the electrode may help to suppress the dissolution of Co at elevated temperature. Evidence is presented suggesting that the oxidation of Co can induce co-oxidation of EDA.

1. Introduction

To enhance the energy density of alkaline batteries, the search for advanced electrode materials has been continuously carried out. Recently, some metal borides, such as Co–B [1–9], Ni–B [10], Fe–B [11,12], V–B and Ti–B [13,14], have been known to own very high discharge capacity in alkaline aqueous solution. For example, VB_2 and TiB_2 electrodes can deliver ultrahigh discharge capacity of 3800 m Ah g^{-1} and 1300 m Ah g^{-1}, respectively. Unfortunately, only Co–B alloy can deliver a considerably reversible discharge capacity. Nanocrystalline Co–B alloy prepared by ball milling method has a reversible discharge capacity of around 238 m Ah g^{-1} [7]. Ultrafine Co–B alloy particles obtained by chemical reduction exhibit better electrochemical properties. Specifically, the alloy can remain the discharge capacity of 300 m Ah g^{-1} after 100 cycles at a high rate of 300 mA g^{-1} [3]. These results show that Co–B alloy may serve as high capacity and high power capability anode for alkaline rechargeable batteries [3]. The electrochemical reaction of Co–B alloy proceeds through the following mechanism [6,11]:

$$CoB_x + OH^- - e \rightarrow Co(OH)_2 + {BO_3}^{3-} + H_2O \quad (1)$$

The reaction with multi-electron charge transfer makes the alloy electrode own high discharge capacity in the first cycles. After B in the alloy is completely oxidized, the whole electrode reaction could be similar to the negative electrode reaction of Ni–Cd battery:

$$Co(OH)_2 + e \underset{\text{discharge}}{\overset{\text{charge}}{\rightleftharpoons}} Co + OH^- \quad (2)$$

B atoms in Co–B alloy electrode serve as an activating agent, which can effectively alleviate the passivation of metal Co and improve further anodic utilization [6]. However, it is known that $Co(OH)_2$ can dissolved into concentrated alkaline solution and its solubility increases dramatically with operating temperature [15,16]. This can result in serious capacity loss of Co–B alloy electrode at elevated temperature and limit the operating temperature range of the electrode. It is widely known that some nitrogen-containing organic compounds are effective corrosion inhibitors for metal materials in various media [17–19]. Our previous study shows that 8-hydroxyquinoline (8-HQ) can drastically enhance the cycle life of Co–B alloy electrode at elevated temperature [20]. The beneficial effect of 8-HQ on the alloy electrode can be attributed to the formation of insoluble complex $(8\text{-HQ})_2Co(II)\,2H_2O$ protective layer that can suppress the dissolution of $Co(OH)_2$

* Corresponding authors at: Institute of Chemistry and Environment, South China Normal University, Guangzhou 510006, China. Tel.: +86 20 39310212; fax: +86 20 39310256.
E-mail address: dhx@scnu.edu.cn (D.S. Lu).

0013-4686/$ – see front matter © 2011 Elsevier Ltd. All rights reserved.
doi:10.1016/j.electacta.2011.02.070

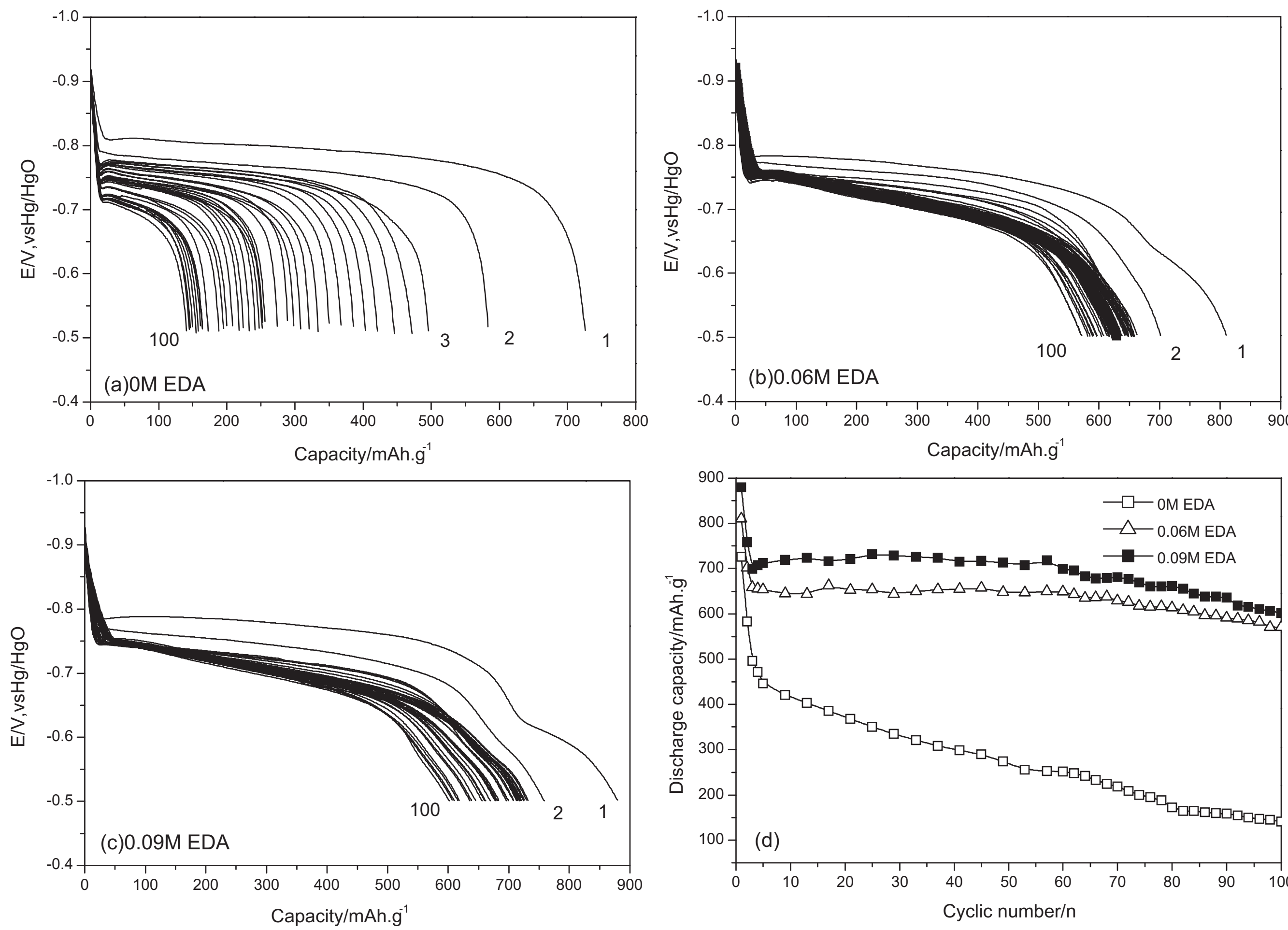

Fig. 1. (a)–(c) Discharge curves and (d) cycle life of Co–B alloy electrode in an electrolyte containing 0 M to 0.09 M EDA at 55 °C.

into electrolyte. In the present study, another nitrogen-containing compound, ethylenediamine (EDA), is added to the electrolyte to suppress the dissolution of $Co(OH)_2$ of Co–B alloy electrode at elevated temperature (55 °C). We found that ethylenediamine can not only improve the cyclability of the electrode, but also greatly enhance its discharge capacity. The reaction mechanism of this electrode system was also proposed.

2. Experimental

2.1. Electrochemical experiment

A Co–B alloy was prepared by chemical reduction process same to our previous study [20]. Chemical composition of the prepared sample is $Co_{57.5}B_{42.5}$ (atomic ratio), determined by using inductivity coupled plasma optical emission spectrum (ICP-OES, Optima2000, Thermo Electron).

Electrochemical experiments were carried out at 55 °C in a traditional three-electrode cell using Solatron1480 battery test system. The working compartment of the cell was separated from the counter electrode compartment by a porous glass frit. The details of the electrochemical cell and the preparation of Co–B alloy electrode were already extensively described in [2]. The prepared electrode consisted of 10 mg 20% Co–B alloy and 80% Ni powder. The base electrolyte was 6 M KOH solution, and the additive electrolyte was 6 M KOH solution containing 0.06 M and 0.09 M of EDA. In the charge–discharge cycle test, the alloy electrodes were first charged at a current of 300 mA g^{-1} for 140 min, and then were discharged at a current of 300 mA g^{-1} to −0.5 V (vs. Hg/HgO). Cyclic voltammetry was performed on the alloy electrode and Ni powder electrode in the electrolyte with 0.09 M EDA. The preparation method and size of the Ni electrode are the same as that of the alloy electrode.

2.2. ICP-OES and Fourier transform infrared spectrometry (FT-IR) experiments

Co content of the cycled electrode was determined by ICP-OES. Surface analysis of the cycled electrode was carried out with Diffuse Reflectance Fourier Transform Infrared Spectrometry (DRFT-IR) (Nicolet 4700). Prior to these measurements, the cycled electrode was thoroughly rinsed with distilled water and vacuum dried at 55 °C for 10 h. To confirm the electrochemical oxidation product of EDA, the cycled electrolyte containing the additive was analyzed by Transmission Fourier Transform Infrared Spectrometry (TFT-IR). Before the measurement, a drop of the electrolyte was added to KBr tablet and the tablet was vacuum dried at 55 °C for 30 min. For comparison, TFT-IR measurement was also performed on the electrolyte before cycling.

3. Results and discussion

Fig. 1 shows the cycle performance of Co–B electrodes in base electrolyte and in the electrolyte containing different concentrations of EDA at 55 °C. In the case of Co–B electrode in EDA – free electrolyte (Fig. 1a), the electrode delivers a high discharge capacity of 723.8 m Ah g^{-1} in the first cycle. However, its discharge capacity

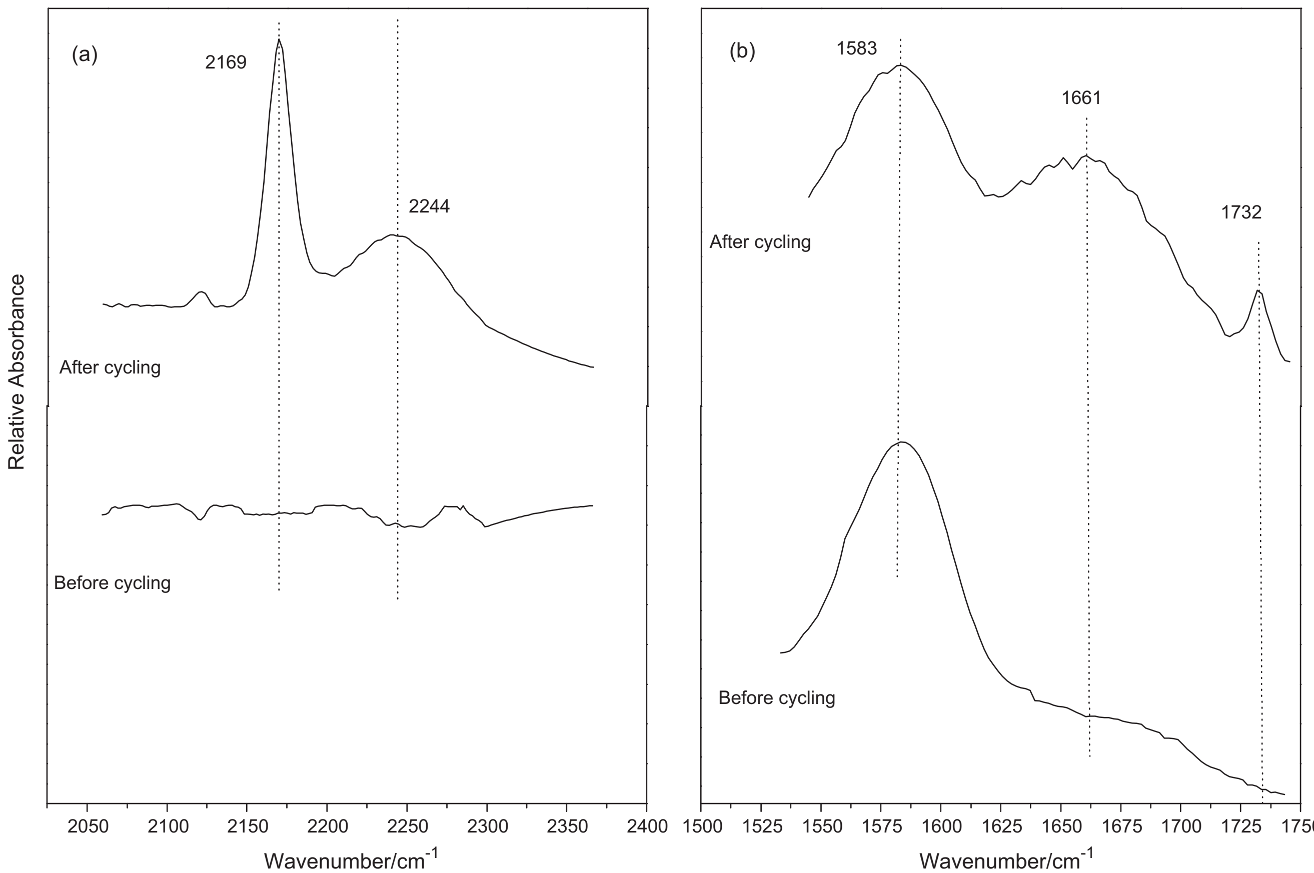

Fig. 2. TFT-IR spectra of the electrolyte with 0.09 M EDA before and after cycling.

is sharply decreased to only 138.5 m Ah g^{-1} after 100 cycles. Cyclability and discharge capacity of Co–B electrode are greatly improved when a small amount of EDA is added into electrolyte (Fig. 1b and c). Specifically, in the electrolytes with 0.06 M and 0.09 M EDA, the discharge capacity of the electrodes after 100 cycles are still up to 570.8 m Ah g^{-1} and 601.7 m Ah g^{-1}, respectively (Fig. 1d). It can be seen that the reversible discharge capacity of the electrode with EDA additive is even almost twice as high as that of the electrode with no additive at room temperature [2,3], indicating that EDA may be oxidized during the discharge cycles and the reaction contributes to part of the discharge capacity. This speculation needs further confirmation.

For the electrode after 100 cycles in based electrolyte (denoted as E_0 electrode), its Co content determined by ICP-OES is 4.37 mg, which is only half of its initial Co content (8.81 mg). However, in the case of the cycled electrodes in the electrolytes containing 0.06 M and 0.09 M EDA (correspondingly denoted as E_1 and E_2 electrodes), their Co content are 7.05 mg and 7.13 mg, respectively. These results show that the serious capacity loss of Co–B alloy electrode at elevated temperature can be attributed to the dissolution of its active substance Co, and EDA additive in the electrolyte can suppress the dissolution of Co.

To identify the electrochemical oxidation product of EDA, TFT-IR spectra of the electrolyte with 0.09 M EDA before and after cycling are shown in Fig. 2. As can be seen from Fig. 2a, in comparison with the electrolyte before cycling, the cycled electrolyte has two absorption bands at 2169 and 2244 cm^{-1}, respectively. The former is assigned to the stretching frequency of the CN^- group [21] and the latter is obviously the C≡N stretching vibration of nitrile [22]. In the 1535–1740 cm^{-1} region (Fig. 2b), only one absorption band appears at 1583 cm^{-1} for the electrolyte before cycling, due to the N–H bending vibration of EDA [23,24]. In the case of the cycled electrolyte, the band located at 1583 cm^{-1} is obviously decreased in intensity, indicating that the concentration of EDA decreases after cycling. In addition, it can be also seen that the cycled electrolyte exhibits two absorption bands at 1661 and 1732 cm^{-1} assigned to the C=O stretching vibration of amide and aldehyde, respectively[25–27]. It can be inferred from above results that EDA can be oxidized during cycling and its oxidation products may contain nitrile and aldehyde, which is good consistent with the result in the literature [28,29]. The electrochemical oxidation of EDA may proceed through the following mechanism:

$$H_2NCH_2CH_2NH_2 - 4e \rightarrow H_2NCH_2CN \quad (3)$$

H_2NCH_2CN is unstable in hot alkaline solution and can be hydrolyzed to amide:

$$H_2NCH_2CN + H_2O \rightarrow H_2NCH_2CONH_2 \quad (4)$$

Part of H_2NCH_2CN may be further oxidized to dicyanogen:

$$H_2NCH_2CN - 4e \rightarrow (CN)_2 \quad (5)$$

Chemical properties of dicyanogen are similar to those of elemental halogen, and it can react with OH^- to produce CN^-:

$$(CN)_2 + OH^- \rightarrow CN^- + CNO^- + H_2O \quad (6)$$

Another part of EDA may be electrochemically oxidized to aldehyde:

$$H_2NCH_2CH_2NH_2 + H_2O - 2e \rightarrow H_2NCH_2CHO + NH_3 \quad (7)$$

Part of H_2NCH_2CHO may be further oxidized:

$$H_2NCH_2CHO + H_2O - 2e \rightarrow HOCCHO + NH_3 \quad (8)$$

Fig. 3 shows typical cyclic voltammograms of Co–B alloy electrode and Ni electrode in the electrolyte with 0.09 M EDA. For

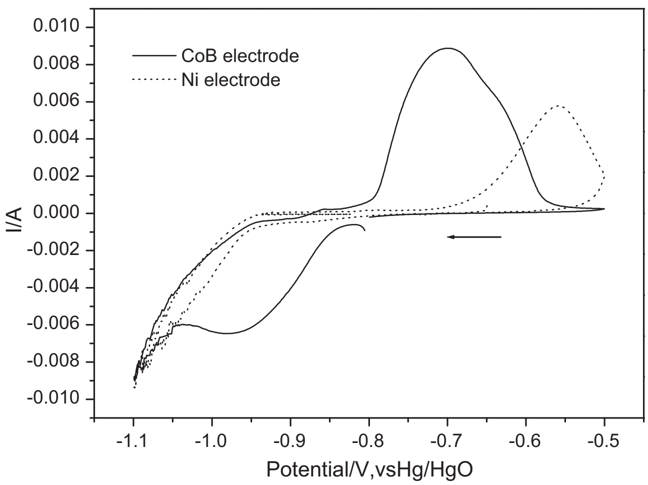

Fig. 3. Typical cyclic voltammograms of Co–B alloy electrode (solid line) and Ni electrode (dotted line) in the electrolyte with 0.09 M EDA. Scan rate: 0.1 mV s^{-1}.

Ni electrode, only one anodic peak at −0.56 V is assigned to irreversible oxidation of EDA. Co–B electrode exhibits a pair of redox peaks, and the reduction peak and oxidation peak appear at −0.97 V and −0.70 V, respectively. Previous studies have shown that the observed redox peaks are attributed to the Co/Co $(OH)_2$ reaction [1,5,6]. However, it can be also seen from Fig. 3 that the current of the oxidation peak is much larger than that of the reduction peak, indicating that electrochemical oxidation of EDA occurs during Co oxidation and contributes to part of oxidation current. This result is consistent with the above IR result. It is worth mentioning that EDA is more easily oxidized on the alloy electrode, suggesting that the oxidation of Co can induce co-oxidation of EDA.

Fig. 4 shows DRFT-IR spectrum of the E_2 electrode. The bands around 1370 and 1482 cm^{-1} are due to the bending vibration of C–H [30], and the two peaks at 2863 and 2935 cm^{-1} are assigned to the stretching vibration of C–H [31]. The C=O stretching vibration of amide can be also observed at 1661 cm^{-1} in this spectrum. In addition, the band at 2122 cm^{-1} may be assigned to cobalt cyanide [32]. These species absorbed on the electrode may play the role of inhibitor and suppress the dissolution of Co in the electrolyte at elevated temperature. However, more studies are needed to understand the exact role of each component in the oxidation products of EDA on the inhibition properties.

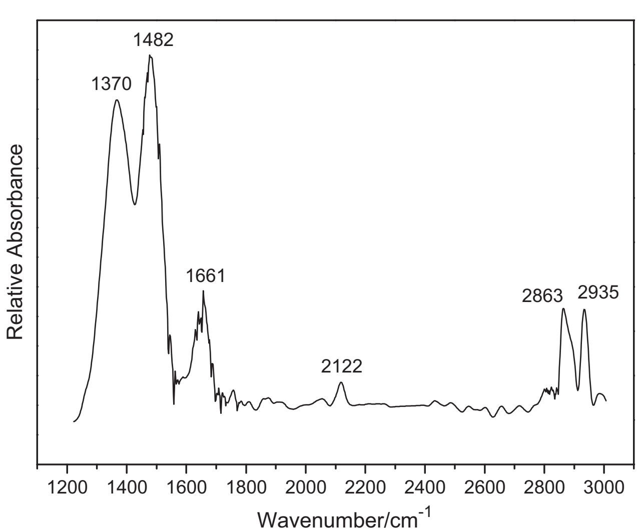

Fig. 4. DRFT-IR spectrum of the E_2 electrode.

4. Conclusions

EDA additive in electrolyte cannot only improve the cyclability of Co–B alloy electrode, but also greatly enhance its discharge capacity at elevated temperature. Specifically, in the electrolyte containing 0.09 M EDA additive, the discharge capacity of the electrode after 100 cycles is still up to 601.7 m Ah g^{-1} at 55 °C. EDA is electrochemically oxidized continuously during discharge cycles with no additional treatment, which contributes to part of the discharge capacity. In addition, the oxidation products absorbed on the electrode may help to suppress the dissolution of Co at elevated temperature. The inorganic–organic hybrid electrode system may have a potential application for electrochemical energy storage.

Acknowledgements

The authors acknowledge financial support by Natural Science Foundation of Guangdong Province (No. 10451063101006346) and Young Teacher Training Plan of Guangdong Universities.

References

[1] D.S. Lu, W.S. Li, X. Jiang, C.L. Tan, R.H. Zeng, J. Alloy Compd. 485 (2009) 621.
[2] D.S. Lu, W.S. Li, C.L. Tan, R.H. Zeng, Electrochim. Acta 55 (2009) 171.
[3] Y.D. Wang, X.P. Ai, H.X. Yang, Chem. Mater. 16 (2004) 5194.
[4] D.W. Song, Y.J. Wang, Y.P. Wang, H.T. Yuan, Electrochem. Commun. 10 (2008) 1486.
[5] Y. Liu, Y.J. Wang, L.L. Xiao, D.W. Song, H.T. Yuan, Electrochem. Commun. 9 (2007) 925.
[6] Y. Liu, Y.J. Wang, D.W. Song, L.F. Jiao, H.T. Yuan, Elecrochim. Acta 53 (2008) 2265.
[7] C. Wu, Y. Bai, X. Wang, F. Wu, C.Z. Zhang, Solid State Ionics 179 (2008) 924.
[8] M. Mitov, A. Popov, I. Dragieva, J. Appl. Electrochem. 29 (1999) 59.
[9] D.G. Tong, Y.Y. Luo, W. Chu, Mater. Chem. Phys. 112 (2008) 907.
[10] J.X. Yu, L. Wang, Y.D. Wang, H. Dong, H.X. Yang, J. Electrochem. Soc. 151 (2008) A1124.
[11] Y.D. Wang, X.P. Ai, Y.L. Cao, H.X. Yang, Electrochem. Commun. 6 (2004) 780.
[12] Y. Bai, C. Wu, F. Wu, L.X. Yang, B.R. Wu, Electrochem. Commun. 11 (2009) 145.
[13] X.W. Yu, S. Licht, Electrochim. Acta 52 (2007) 8138.
[14] X.W. Yu, S. Licht, J. Power Sources 179 (2008) 407.
[15] P. Elumalai, H.N. Vasan, N. Munichandraiah, J. Power Sources 93 (2001) 201.
[16] F. Tronel, L. Guerlou, L. Goubault, P. Bernard, C. Delmas, J. Power Sources 179 (2008) 837.
[17] S.V. Lamaka, M.L. Zheludkevich, K.A. Yasakau, M.F. Montemor, M.G.S. Ferreira, Electrochim. Acta 52 (2007) 7231.
[18] D. Gallant, M. Pézolet, S. Simard, Electrochim. Acta 52 (2007) 4927.
[19] L. Wang, J.X. Pu, H.C. Luo, Corros. Sci. 45 (2003) 677.
[20] D.S. Lu, W.S. Li, F.M. Xiao, R.H. Tang, Electrochem. Commun. 12 (2010) 362.
[21] C.M. Pharr, P.R. Griffiths, Anal. Chem. 69 (1997) 4673.
[22] Z. Getahun, C.Y. Huang, W.F. DeGrado, F. Gai, J. Am. Chem. Soc. 125 (2003) 405.
[23] S.H. Brewer, A.M. Allen, S.E. Lappi, T.L. Chasse, K.A. Briggman, C.B. Gorman, S. Franzen, Langmuir 20 (2004) 5512.
[24] Y. Hirashima, H. Sato, A. Suzuki, Macromolecules 38 (2005) 9280.
[25] J. Oomens, N. Polfer, D.T. Moore, L. Meer, A.G. Marshall, J.R. Eyler, G. Meijere, G. Helden, Phys. Chem. Chem. Phys. 7 (2005) 1345.
[26] B. Yan, Q. Sun, J.R. Wareing, C.F. Jewell, J. Org. Chem. 61 (1996) 8765.
[27] H. Sano, H. Maeda, T. Ichii, K. Murase, K. Noda, K. Matsushige, H. Sugimura, Langmuir 25 (2009) 5516.
[28] M. Fleischmann, K. Korinek, D. Pletcher, J. Electroanal. Chem. 31 (1971) 39.
[29] N.A. Hampson, J.B. Lee, J.R. Morley, B. Scanlon, Can. J. Chem. 47 (1969) 3729.
[30] P.J.R. Honeybone, R.J. Newport, J.K. Walters, Phys. Rev. B 50 (1994) 839.
[31] G.W. Gokel, Dean's Handbook of Organic Chemistry, 2nd ed., McGraw-Hill, NY, 2006.
[32] R. Martínez-Garcí, M. Knobelb, G. Goyac, M.C. Gimenezd, F.M. Romerod, E. Reguera, J. Phys. Chem. Solids 67 (2006) 2289.